

Simulation Input & Data Standard - (SIDS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Simulation Input & Data Standard (SIDS) defines the governance conditions under which data, values, signals, measurements, states, distributions, records, synthetic inputs, externally supplied information, and other simulation inputs are identified, qualified for simulation use, assessed for representativeness, transformed under controlled conditions, temporally aligned, uncertainty-aware, and bound to the Simulation Object, Purpose & Scope, Reality Representation, Assumptions, Model, Environment, and applicable simulation configuration.

SIDS establishes the governed input boundary through which information becomes capable of materially influencing simulation behavior.

It ensures that information does not become simulation input merely because it is technically available or technically ingestible.

Core Governance Question

Which inputs and data may enter the simulation, and under what representativeness, transformation, temporal, and uncertainty conditions?

Purpose

A simulation may be structurally well designed and correctly configured while still producing misleading behavior because the information entering it is: irrelevant, unrepresentative, temporally misaligned, incorrectly transformed, incompletely sourced, improperly aggregated, synthetically generated without disclosure, stale, overly precise, uncertain, or inconsistent with the simulation conditions it is intended to represent.

SIDS exists to govern that problem.

The standard separates:

having data

from:

having data suitable to enter this simulation under these conditions.

Operational Role

SIDS is the sixth Parent Standard of SIMULOS® — Simulation Governance Architecture.

The preceding standards establish:

- 1. Simulation Object — What exactly is being simulated?**
- 2. Simulation Purpose & Scope — Why is it being simulated and what must it cover?**
- 3. Reality Representation — Which aspects of reality must be represented and how?**
- 4. Simulation Assumptions — Which premises are required and what limitations do they introduce?**
- 5. Simulation Model & Environment — Through which controlled model and environment will the simulation operate?**

SIDS asks:

Which information may now enter that governed model and environment, and under what conditions is that information suitable to shape simulation behavior?

It therefore establishes the input basis required before governed Scenario Design.

Governance Flow

Governed Simulation Model & Environment

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Input Need

↓

Simulation Input Identification

↓

Representativeness Assessment

↓

Transformation Governance

↓

Temporal Qualification

↓

Input Uncertainty

↓

Input Acceptance / Restriction / Exclusion



Governed Simulation Input & Data Basis



Simulation Scenario Design

Operational Space

SIDS governs simulation-specific conditions surrounding: input identification, input classification, input origin, input provenance references, data representativeness, population representativeness, environmental representativeness, state representativeness, distributional representativeness, temporal representativeness, input transformation, normalization, conversion, aggregation, disaggregation, interpolation, extrapolation, filtering, encoding, synthetic-data use, derived inputs, temporal alignment, timestamp relevance, input currency, temporal windows, sequence, synchronization, uncertainty attached to input, input ranges, missing information, conflicting inputs, input substitution, input restrictions, and input-to-simulation traceability.

Operational Scope

SIDS applies only where data or other information becomes an input into a simulation.

It may govern inputs originating from: sensors, operational systems, historical records, scientific measurements, engineering systems, external databases, third-party providers, human operators, AI systems, autonomous systems, previous simulations, synthetic-data generators, probabilistic models, statistical models, digital twins, live operational feeds, batch datasets, scenario specifications, or manually supplied values.

SIDS is not a general-purpose Data Governance Architecture.

It does not attempt to govern the complete lifecycle of data across an organization.

Its boundary is narrower:

the point at which information is proposed for use as simulation input and the conditions under which that use remains methodologically governable.

Simulation Input

A Simulation Input is any governed information, value, state, signal, event, distribution, condition, record, or data object introduced into the simulation and capable of influencing its configuration, behavior, progression, or outputs.

Simulation Inputs may include: measured values, observed states, environmental values, actor states, initial conditions, time-series data, event streams, probability distributions, behavioral data, control signals, geographic information, scientific measurements, operational records, system telemetry,

or synthetic values.

Input ≠ Parameter

SIDS distinguishes Simulation Input from Simulation Parameter.

A parameter configures the model or environment.

An input supplies information consumed by the simulation.

For example:

Vehicle Mass = 1,800 kg

may be a parameter.

Current Vehicle Position = X, Y

may be an input.

The distinction should follow simulation function rather than software implementation. Input ≠ Assumption

A Simulation Assumption establishes a premise.

A Simulation Input supplies information.

For example:

Assumption: Network latency remains within normal operating range.

Input: Current network latency = 47 ms.

An assumption may influence interpretation of an input, but the governance objects remain distinct. Input ≠ Evidence by Default

A Simulation Input may later contribute to simulation evidence.

However:

Input into a simulation is not automatically validation evidence merely because the simulation consumed it.

VALIDOS® governs formal Validation Evidence Fitness where such evidence is used for validation.

Input Identification

Every material Simulation Input should be sufficiently identifiable.

Where relevant, identification should establish: input identity, input type, source reference, applicable Simulation Object, applicable model or environment element, temporal context, unit or representation, transformation status, uncertainty status, and simulation configuration.

Input Classification

Simulation Inputs may be classified as:

Observed

Measured

Derived

Estimated

Synthetic

Historical

Live

External

Human-Supplied

Model-Generated

Scenario-Specified

or another simulation-specific class.

Classification supports different governance treatment without assuming one class is universally superior.

Input Origin

Input Origin identifies where the Simulation Input came from.

The origin may include: instrument, system, dataset, organization, individual, model, simulator, external service, AI system, or synthetic-generation process.

SIDS preserves origin visibility.

It does not replace general Source Reliability governance under VALIDOS® where formal source qualification is required. Representativeness

A Simulation Input may be factually correct yet still be unsuitable for the simulation because it does not adequately represent the relevant simulation conditions.

SIDS therefore governs Simulation Data Representativeness.

Representativeness asks:

Does this input sufficiently represent the population, environment, state, distribution, time, behavior, or condition that the simulation intends to model?

Representativeness Is Purpose-Relative

No dataset is universally representative.

Representativeness depends upon:

Simulation Object + Purpose + Scope + Reality Representation + Scenario Context

A dataset suitable for one simulation inquiry may be materially unsuitable for another.

Population Representativeness

Where simulation data represents humans, systems, assets, organizations, biological entities, or other populations, SIDS asks whether the input captures materially relevant population characteristics.

This may include: subgroup differences, distribution differences, behavior, demographic characteristics, technical classes, system variants, or operating profiles.

Environmental Representativeness

Input data may represent only certain environments.

For example:

dry-road vehicle data

may not sufficiently represent:

snow-covered road conditions.

Environmental applicability should remain explicit.

Operational-State Representativeness

Data from normal operation may not represent: degraded operation, failure, emergency states, recovery states, or extreme conditions.

The operating state from which input data originates should therefore remain relevant to simulation-input governance.

Distributional Representativeness

Average values may conceal: variance, tails, multimodality, extreme events, correlations, clusters, or rare conditions.

Where these properties materially affect simulation behavior, SIDS requires their representativeness to remain visible.

Representativeness Gap

A Simulation Data Representativeness Gap exists where the available input data does not sufficiently represent a material part of the simulation inquiry.

A gap may be:

Non-Material

Material but Bounded

Conditional

Material and Unresolved

or:

Representativeness Undetermined Input Transformation

Simulation data frequently cannot enter a simulation in its original form.

It may require: cleaning, filtering, normalization, scaling, unit conversion, coordinate conversion, encoding, aggregation, interpolation, extrapolation, feature derivation, compression, imputation, or another transformation.

SIDS requires material transformations to remain visible.

Transformation Principle

SIMULOS® establishes:

A transformed simulation input is not methodologically identical to its source data merely because the transformation is technically routine.

Transformations may: remove information, introduce assumptions, create synthetic values, change distributions, alter temporal meaning, suppress outliers, or reshape dependencies.

Transformation Traceability

Where material:

Source Input → Transformation → Transformed Input → Simulation Use

should remain reconstructable.

Transformation Loss

A transformation may remove information.

Examples include: aggregation, compression, filtering, dimensional reduction, categorization, or rounding.

Material information loss should remain visible where it can affect simulation behavior. Transformation Gain ≠ New Reality

Interpolation, extrapolation, synthetic extension, or statistical reconstruction may create additional values.

SIMULOS® establishes:

Generated or inferred values must not silently become treated as directly observed reality.

Synthetic Simulation Input

A Synthetic Simulation Input is generated rather than directly observed from the represented reality.

Synthetic inputs may be legitimate and highly useful.

Their synthetic origin must remain identifiable.

Examples include: synthetic populations, generated weather, simulated sensor streams, artificial failure sequences, generated network traffic, synthetic user behavior, or AI-generated input data.

Synthetic ≠ Inferior by Default

SIDS does not assume synthetic input is methodologically weaker.

It requires that its role and generation basis remain visible.

In some simulation contexts synthetic data may be required specifically to test conditions that real-world datasets do not sufficiently contain.

Derived Input

A Derived Input is calculated from one or more existing data objects.

Conceptually:

Input A + Input B + Derivation Method → Derived Input C

The derivation should remain traceable where C materially affects the simulation.

Temporal Data Governance

Simulation data has a temporal relationship to the reality it represents.

SIDS governs: when the data was generated, when it was observed, which period it represents, whether it remains current, whether sequence matters, whether synchronization is required, and whether temporal aggregation changes meaning.

Data Currency

Recent data is not automatically better data.

Historical data is not automatically unsuitable.

The correct question is:

Is the temporal context of this input appropriate for the simulation inquiry?

Temporal Applicability

An input should remain connected to the period or temporal condition it represents where material.

Data from:

System Version 1 in 2022

may not represent:

System Version 4 in 2026.

Temporal Alignment

Multiple input streams may originate from different times.

Where the simulation assumes they represent the same state, temporal alignment must be sufficiently governed.

Temporal Misalignment

A Temporal Misalignment exists where inputs treated as simultaneous or causally connected actually represent materially different time states.

This can create simulation behavior that never existed in the represented reality.

Temporal Sequence

Some simulation behavior depends not only upon values but upon their order.

SIDS therefore preserves sequence where:

$A \rightarrow B \rightarrow C$

is materially different from:

$C \rightarrow A \rightarrow B$.

Temporal Aggregation

Data aggregated across long periods may hide: transient events, peaks, seasonality, failures, transitions, or rapid variability.

Where these phenomena matter, temporal aggregation may create an input

limitation.

Input Uncertainty

A Simulation Input may contain uncertainty arising from: measurement, estimation, missing values, transformation, sampling, temporal mismatch, synthetic generation, incomplete representativeness, sensor limits, model derivation, or another source.

SIDS requires material input uncertainty to remain visible. Input Uncertainty $\neq$ Simulation Assumption Uncertainty

SAUM governs uncertainty attached to assumptions.

SIDS governs uncertainty attached to simulation inputs and data.

The two may interact but remain distinct governance spaces.

Measurement Uncertainty

Measured values may include: tolerance, error range, confidence interval, noise, or calibration limitation.

Simulation use should not imply greater certainty than the measurement supports.

Estimated Input

Estimated inputs may be necessary where direct measurement is unavailable.

Where material, the simulation should distinguish:

Measured Input

from:

Estimated Input.

Missing Input

A Missing Simulation Input exists where the simulation requires information that is not available.

Missing data should not silently become zero, average, default, or synthetic replacement without an identifiable treatment.

Missing Input Treatment

Possible treatments may include:

Exclude

Impute

Substitute

Generate Synthetic Input

Restrict Scenario

Proceed Conditionally

or:

Input Unresolved

depending upon the simulation context.

No Missing-to-Zero Rule

SIMULOS® establishes:

Missing information must not automatically be interpreted as a zero value unless zero is itself a governed and justified input.

Conflicting Inputs

Multiple input sources may provide materially inconsistent values.

For example:

Sensor A = 50

Sensor B = 75

Where both are materially credible, the conflict should remain visible.

SIDS does not require one universal conflict-resolution method.

It requires the resolution basis to remain governable.

Input Precedence

Where multiple data sources exist, the simulation may establish rules determining which source takes precedence.

Such precedence should not be arbitrary where it materially affects simulation behavior.

Input Substitution

A planned Simulation Input may be replaced by another.

Example:

Live Sensor Feed → Historical Replay Data

or:

Measured Dataset → Synthetic Dataset

Material substitution should remain visible.

Input Equivalence

Two input sources should not be treated as equivalent merely because they share the same variable name or data format.

Equivalence depends upon materially relevant: meaning, population, environment, time, precision, transformation, uncertainty, and origin.

Input Compatibility

Simulation Inputs should remain compatible with: Simulation Model, Simulation Environment, parameter configuration, units, temporal structure, spatial structure, expected data type, state representation, and applicable scenario conditions.

Technical ingestion alone does not establish compatibility.

No Technical Ingestibility Principle

SIMULOS® establishes:

The fact that a simulation can ingest data does not establish that the data is suitable to influence the simulation.

Input Acceptance

A proposed input may receive a simulation-specific status such as:

Accepted

Conditionally Accepted

Accepted with Limitation

Rejected

Substituted

or:

Input Status Undetermined

The status applies to its role within the simulation.

It is not a universal judgment about the data itself.

Input Restriction

A Simulation Input may be suitable only for defined: scenarios, environments, populations, model versions, temporal periods, geographic regions, system states, or analysis purposes.

Restrictions should remain bound to the affected input.

Input-to-Model Binding

Material inputs should remain traceable to the model components they influence.

Conceptually:

Simulation Input → Model Component → Simulation Behavior
Input-to-Environment Binding

Some inputs configure or update the Simulation Environment.

Conceptually:

Simulation Input → Environmental Element / State

This relationship should remain visible where material.

Input-to-Scenario Binding

Some inputs may apply only within particular scenarios.

SIDS preserves the input relationship.

SSS governs scenario design itself.

Input-to-Execution Binding

A simulation execution should be capable of identifying the materially relevant input set it consumed.

Conceptually:

Simulation Execution ID → Simulation Input Set ID Simulation Input Set

A Simulation Input Set is the governed collection of materially relevant inputs used for a defined simulation execution or scenario.

The set may identify: input identities, values, origin, transformation state, temporal state, uncertainty, restrictions, and applicable execution.

Input Set Versioning

Where materially different input sets can produce materially different behavior, they should remain distinguishable.

A later update to input data should not silently rewrite the historical input basis of a previous execution.

Point-in-Time Input Reconstruction

SIDS should support questions such as:

Which data entered this simulation run?

What did those inputs represent?

What transformations had occurred?

Which time periods did they represent?

What uncertainty was attached?

Which inputs were synthetic or estimated?

Which inputs were missing, substituted, or restricted?

No Input Laundering Principle

SIMULOS® establishes:

A simulation input must not acquire greater representativeness, certainty, currency, or evidentiary strength merely because it has passed through a simulation.

No Representativeness Inflation Principle

A dataset representative of:

Population A

must not silently become representative of:

Population A + B + C

simply because the simulation contains all three populations.

No Temporal Inflation Principle

Historical data must not silently become current-state data merely because it is replayed inside a current simulation environment.

No Transformation Concealment Principle

A materially transformed input should not be represented as if it were the original data object.

No Synthetic-to-Observed Conversion

Principle

Synthetic data must not become labeled or interpreted as observed reality merely because simulation execution used it successfully.

Input Change

Simulation input conditions may change when: new data arrives, source systems change, transformations change, temporal windows change, data is corrected, additional populations are included, uncertainty changes, or new representativeness gaps are discovered.

Material changes should remain traceable.

Input Change Propagation

A material input change may affect:

Scenario Design → Simulation Execution → Simulation Output → Reality
Correlation → downstream Simulation Evidence

SIDS identifies the input change.

Subsequent Parent Standards govern their respective downstream consequences.

Input Drift

Input Drift occurs where the statistical, behavioral, environmental, temporal, or semantic characteristics of incoming Simulation Inputs materially change relative to the input basis under which the simulation was designed.

Drift may be: population drift, distribution drift, environment drift, temporal drift, source drift, measurement drift, or semantic drift.

SIDS governs drift at the simulation-input boundary.

Broader lifecycle drift and resimulation governance remain with the later SIMULOS® lifecycle space.

Input Reassessment Trigger

Material changes in: Simulation Object, Purpose, Scope, Reality Representation, Assumptions, Model, Environment, scenario context, source data, transformation process, or time

may require input reassessment.

Governed Simulation Input & Data Basis

At completion of SIDS, the simulation should possess a sufficiently governed Simulation Input & Data Basis establishing: which inputs may enter, what they represent, where they originate, how they were transformed, which temporal conditions apply, what uncertainty remains, where restrictions exist, and which simulation configuration may use them.

Governance Boundary

SIDS begins when a governed Simulation Model & Environment requires information, data, states, values, signals, or other inputs capable of influencing simulation activity.

SIDS ends when the materially relevant inputs have been sufficiently identified, assessed for representativeness, transformation-governed, temporally qualified, uncertainty-aware, restricted where necessary, and assembled into a governed Simulation Input & Data Basis suitable for Simulation Scenario Design.

Architectural Boundary

SIDS does not govern: enterprise-wide data governance, universal data quality, general database governance, complete data lifecycle management, general source reliability, formal validation of data, model parameterization, Simulation Scenario Design, Simulation Execution, Simulation Output, or operational reliance.

Where those questions arise, the appropriate SIMULOS® Parent Standard or complementary OOF® architecture applies.

Minimum Governance Requirements

A conforming application of SIDS should establish:

1. identifiable material Simulation Inputs, 2. input classification and origin, 3. simulation-specific representativeness, 4. material representativeness gaps, 5. explicit material transformations, 6. transformation traceability, 7. synthetic and derived input identification, 8. temporal context and currency, 9. temporal alignment where required, 10. material input uncertainty, 11. missing-input treatment, 12. conflict and substitution treatment, 13. input restrictions and applicability, 14. input-to-model/environment binding, 15. input-to-execution binding, 16. controlled Input Sets, 17. material input-change visibility, 18. and sufficient Input & Data traceability for reconstruction.

Governance Outputs

SIDS may produce: Simulation Input Register, Simulation Input Identifier, Simulation Input Classification, Simulation Input Origin Record, Simulation Input Set, Simulation Input Set Identifier, Input-to-Model Binding, Input-to-Environment Binding, Input-to-Scenario Reference, Input-to-Execution Binding, Simulation Data Representativeness Assessment, Population Representativeness Record, Environmental Representativeness Record, Operational-State Representativeness Record, Distributional Representativeness Record, Representativeness Gap Record, Simulation Data Transformation Record, Transformation Traceability Map, Transformation Loss Record, Synthetic Simulation Input Record, Derived Input Record, Temporal Data Record, Data

Currency Assessment, Temporal Applicability Record, Temporal Alignment Record, Temporal Misalignment Record, Simulation Input Uncertainty Record, Measurement Uncertainty Record, Missing Input Record, Missing Input Treatment Record, Conflicting Input Record, Input Precedence Record, Input Substitution Record, Input Compatibility Assessment, Simulation Input Acceptance Status, Input Restriction Record, Input Change Record, Input Drift Record, Input Reassessment Trigger, Point-in-Time Input Reconstruction Record, Simulation Input & Data Traceability Map, and Governed Simulation Input & Data Basis.

Relationship to Other OOF® Architectures

SIDS operates specifically at the simulation-input boundary.

VALIDOS® may govern formal validation of input datasets, their evidence fitness, source reliability, or claims made from them.

ORA™ may establish actual Operational Reality from which live or historical inputs originate.

INTEGROS® may govern integrity of input objects, transformations, records, and preserved input states.

ADIT® may preserve evidence concerning input acquisition, transformation, changes, and execution use.

AGA™ may govern accountability for material input-selection or substitution decisions.

Other OOF® architectures may govern the objects represented by the input.

SIDS does not replace those architectures.

It governs specifically:

whether and under what simulation-specific conditions information may enter and influence a governed simulation.

Architectural Position

Simulation Input & Data Standard (SIDS) is the sixth Parent Standard of SIMULOS® — Simulation Governance Architecture.

Its position in the 12-standard architecture is:

1. Simulation Object

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2. Simulation Purpose & Scope

↓

3. Reality Representation

↓

4. Simulation Assumptions



5. Simulation Model & Environment



6. Simulation Inputs & Data



7. Simulation Scenario Design



8–12. Downstream SIMULOS® lifecycle governance

SIDS therefore sits at the transition between:

a controlled simulation structure

and:

the deliberate construction of what conditions that structure will be asked to simulate.

The Model & Environment determine where and how simulation can occur.

SIDS determines what information may enter it.

The next Parent Standard, Simulation Scenario Standard (SSS), determines which governed combinations of conditions must actually be simulated.

Foundational Principle

Information becomes legitimate simulation input only when what it represents, how it was transformed, when it applies, what uncertainty it carries, and where it may influence the simulation remain sufficiently governed.

Canonical Closing Statement

Simulation Input & Data Standard establishes the governed information boundary through which data, values, signals, states, measurements, distributions, synthetic information, and other simulation inputs enter SIMULOS®. By governing input identification, representativeness, transformation, temporal applicability, uncertainty, missing and conflicting information, substitution, restrictions, execution binding, change, drift, and reconstruction, SIDS prevents technically ingestible information from being mistaken for methodologically suitable simulation input and ensures that every materially influential input remains connected to the conditions under which it may legitimately shape simulation behavior.

Module Architecture

→ [Simulation Input Identification Module \(SIIM\)](#).

→ [Simulation Data Representativeness Module \(SDRM\)](#).

→ [Simulation Data Transformation Module \(SDTM\)](#).

→ [Simulation Temporal Data Module \(STDM\)](#).

→ [Simulation Input Uncertainty Module \(SIUM\)](#).

Related Documents

→ [About Simulation Input & Data Standard \(SIDS\)](#).

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